Engineering Education in the Age of ChatGPT and Exponential AI:

A Student Primer

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QATAR UNIVERSITY

Outline

1. ChatGPT/ Generative AI and its promise for education
2. A Critical Look at ChatGPT
3. Education and Learning: A Primer
4. How to Effectively Use ChatGPT/ AI

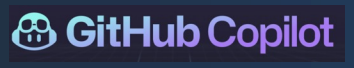
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A brief intro to ChatGPT/ Generative AI and its promise for education

Text Generator



Code Generator



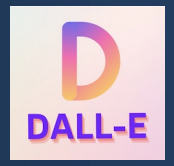


Image Generator



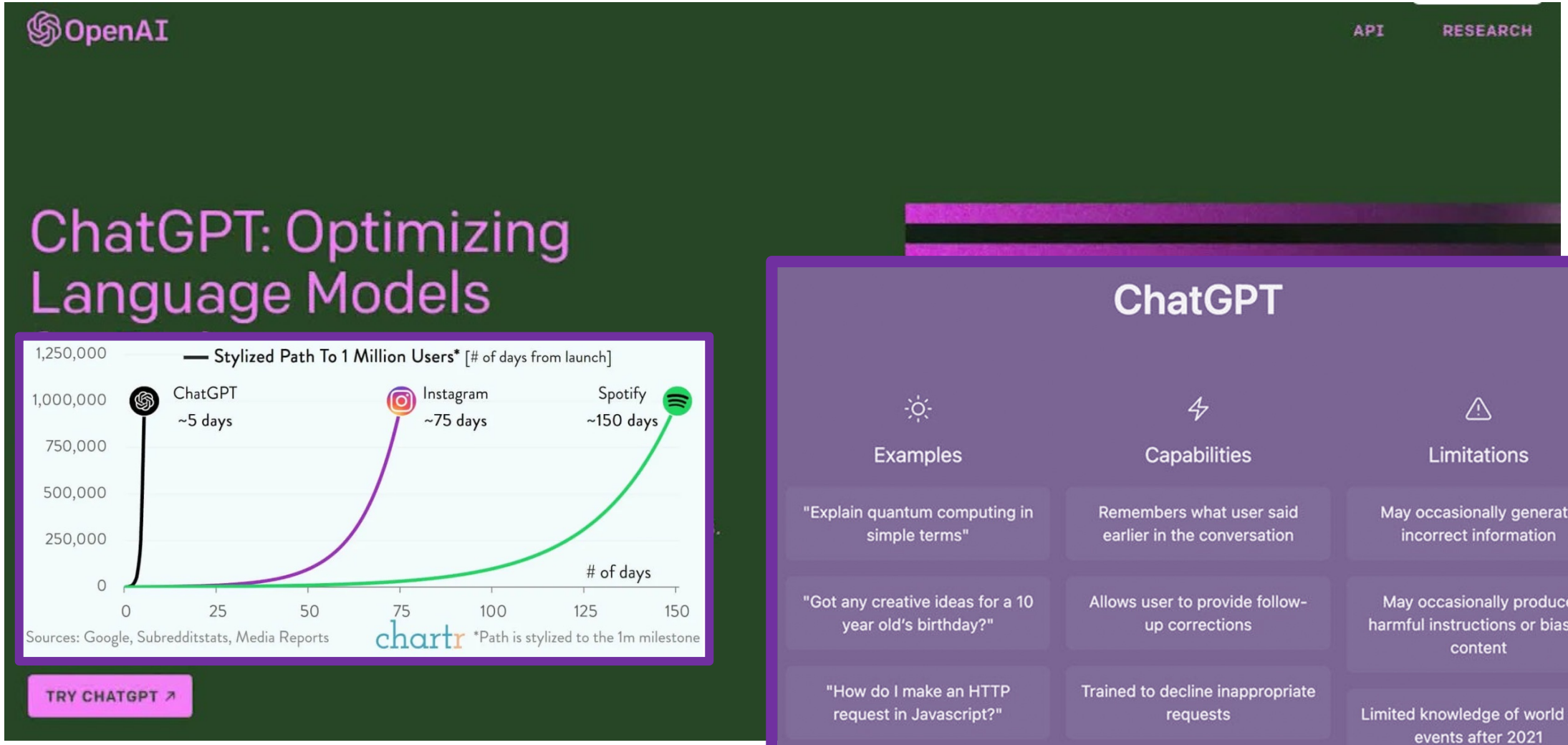
Speech Synthesizer



Conversational AI (ChatBot)

About ChatGPT

https://openai.com/blog/chatgpt/



CODE GENERATION

JU Give me code for the Dijkstra algorithm in Python

Here is the code for the Dijkstra algorithm in Python:

Copy code

```python
from collections import defaultdict
import heapq

def dijkstra(graph, source):
  # Initialize distances and predecessor for each vertex
  distances = defaultdict(lambda: float('inf'))
  predecessor = defaultdict(lambda: None)
  distances[source] = 0

  # Initialize priority queue with all vertices and their distances from the source
  priority_queue = [(0, source)]
  while priority_queue:
    # Extract vertex with minimum distance from the queue
    distance, vertex = heapq.heappop(priority_queue)

    # Update distances and predecessors for all neighbors of the vertex
    for neighbor, weight in graph[vertex]:
      if distances[neighbor] > distance + weight:
        distances[neighbor] = distance + weight
        predecessor[neighbor] = vertex
        heapq.heappush(priority_queue, (distances[neighbor], neighbor))

  return distances, predecessor
```

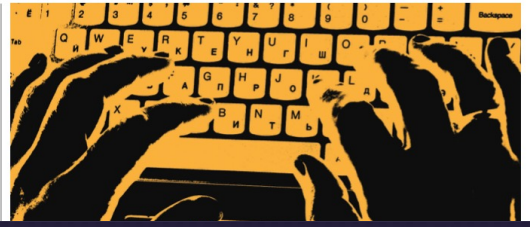
DOI:10.1145/3570220

Matt Welsh

Viewpoint

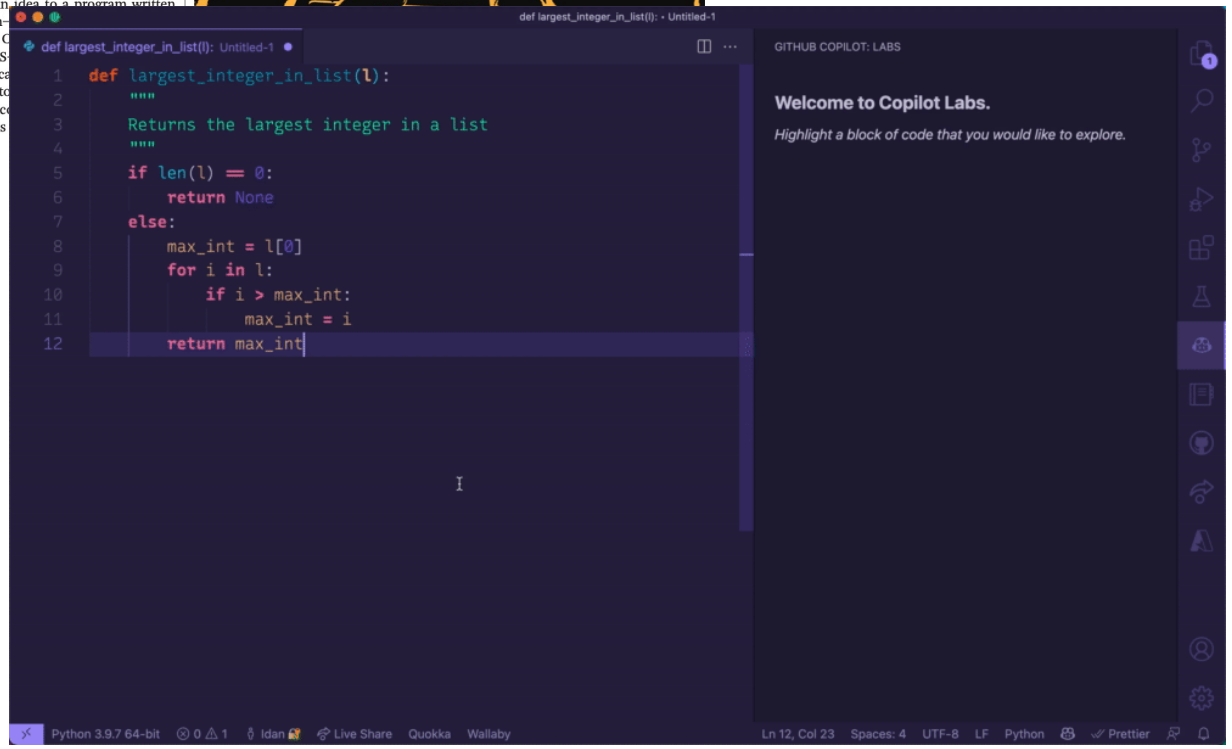
The End of Programming

The end of classical computer science is coming, and most of us are dinosaurs waiting for the meteor to hit.



I CAME OF AGE in the 1980s, programming personal computers such as the Commodore VIC-20 and Apple][e at home. Going on to study computer science (CS) in college and ultimately getting a Ph.D. at Berkeley, the bulk of my professional training was rooted in what I will call "classical" CS: programming, algorithms, data structures, systems, programming languages. In Classical Computer Science, the ultimate goal is to reduce an idea to a program written by a human—like Java or C. Classical CS or sophistica algorithm to tuse Paxos c expressed as

GitHub Copilot



Personalized Learning

The Two-Sigma Effect of Individualized Tutoring



*Students learn best when they receive **formative feedback** about specific qualities of their work along with concrete advice on what they can do to improve.*

VIRTUAL TUTORING & PERSONALIZED LEARNING

JU

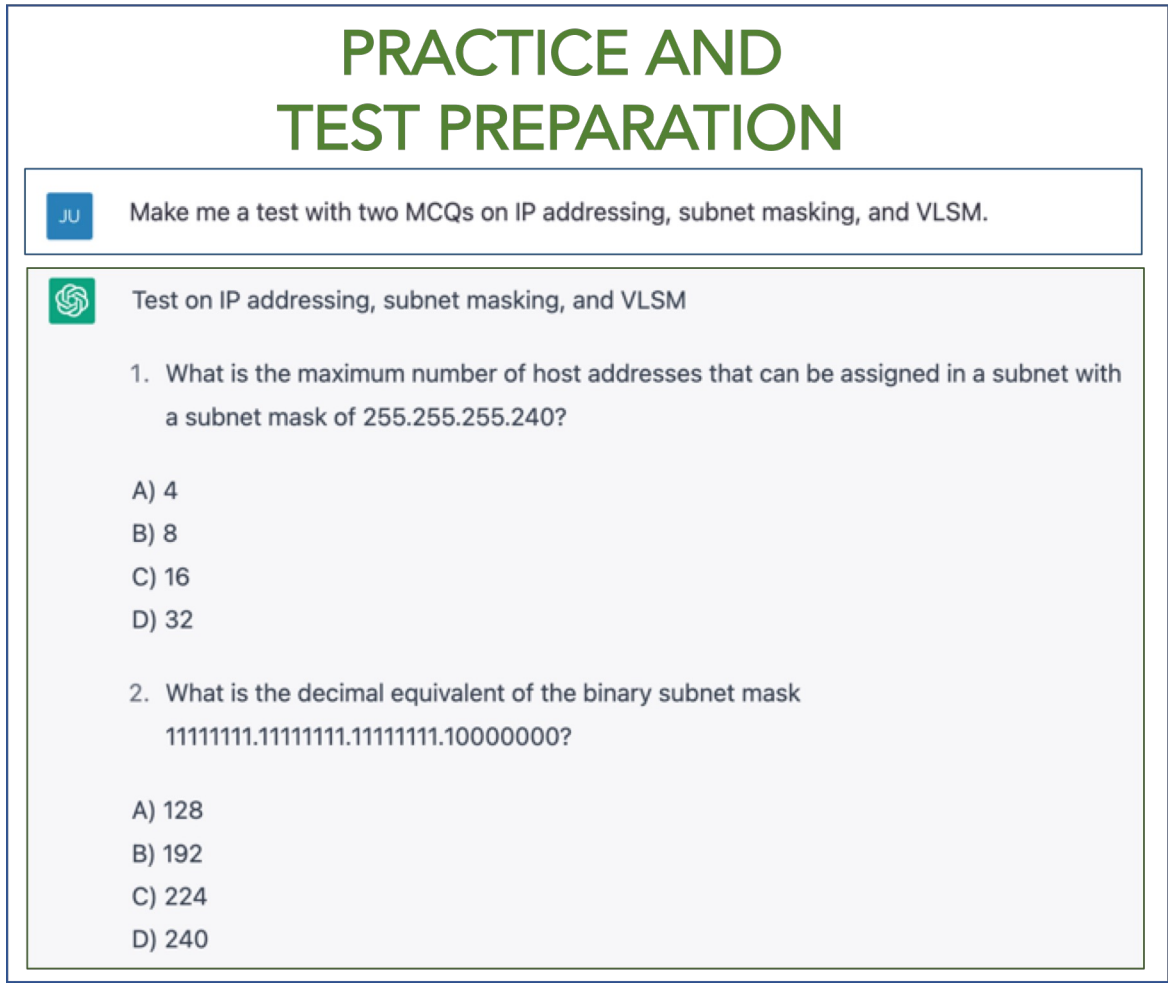
My teacher asked me, "What is the maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240?". My reply was 8 but my teacher said that this is wrong. Why is it wrong?

The maximum number of host addresses that can be assigned in a subnet with a subnet mask of 255.255.255.240 is not 8. The subnet mask 255.255.255.240 allows for 4 bits to be used for host addresses, and each bit can represent 2 values (0 or 1). Therefore, the total number of host addresses that can be assigned in this subnet is 2^4 = 16.

It's important to note that the subnet mask is used to determine the number of host addresses that can be assigned in a subnet. The subnet mask consists of a series of 1s and 0s, and the number of 1s indicates the number of bits that are reserved for the network portion of the IP address. The number of 0s indicates the number of bits that are available for host addresses. By counting the number of 0s in the subnet mask and using the formula 2^n (where n is the number of 0s), you can determine the maximum number of host addresses that can be assigned in a subnet.

Test Preparation

"In virtually all areas of learning, you build better mastery when you use testing as a tool to identify and bring up your areas of weakness."—Brown, Roediger, and McDaniel



Services for Writers/ Researchers/ Essayists

LANGUAGE LEARNING, EDITING, AND WRITING

Edit the following for me: "An important consideration regarding the use of automated thinking and writing tools is that how will it impact the livelihoods of those who are in the business of writing." and provide feedback if it was good writing or if it contained any mistakes or issues.

Here is a revised version of the sentence:

"One important consideration when using automated thinking and writing tools is how they will affect the livelihoods of those who make their living as writers."

This revised version is clearer and more concise than the original sentence. I did not identify any mistakes or issues in the original sentence, but I made a few changes to improve its clarity and flow. Specifically, I replaced the phrase "regarding the use of" with the more concise "when using," and I rephrased the final part of the sentence to be more concise and to use more specific language. Overall, the revised sentence is a well-written and clear expression of the idea that the use of automated tools could have an impact on the livelihoods of writers.

STORY WRITING AND PERSUASIVE WRITING

JU Can you tell me a nice story of how too much use of a technology can actually be detrimental

Sure, here is a story about the negative consequences of overusing a technology:

There was once a small village nestled in the mountains, where the people lived a simple and self-sufficient life. They grew their own food, raised animals, and spent their days working and playing in the great outdoors.

One day, a group of traveling merchants arrived in the village, bringing with them a new technology that promised to revolutionize the way the villagers lived. It was a machine that could do the work of many people, and the merchants convinced the villagers to buy it, promising that it would make their lives easier and more efficient.

At first, the villagers were excited about the new technology. They used it to plow their fields, churn their butter, and even make their meals. But as time passed, they found that they were relying on the machine more and more, and they began to neglect their own skills and abilities. They stopped gardening, stopped caring for their animals, and stopped doing many of the tasks that had once been a part of their daily lives.

As the villagers became more and more dependent on the machine, they started to lose their sense of community and connection with each other. They spent more time indoors, sitting in front of screens and interacting with the machine, and less time outdoors, working and playing together.

One day, the machine broke down, and the villagers realized that they had lost the knowledge and skills they needed to repair it. They were completely reliant on the machine, and without it, they were lost. They had to rely on the merchants to fix it, and they had to pay a high price for the repair.

The villagers learned a valuable lesson that day: that too much reliance on a single technology can be detrimental, and that it is important to maintain their own skills and abilities, and to nurture their sense of community and connection with each other.

Mathematics Questions

Find the derivative of the function f(x) = 3x^2 + 2x - 1. Describe your working

To find the derivative of the function f(x) = 3x^2 + 2x - 1, we need to use the rules of differentiation. The derivative of a function is a measure of how the function is changing at a particular point.

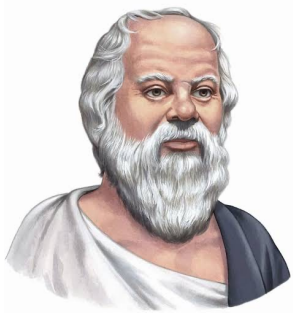
The derivative of a constant term (such as -1 in this case) is always 0, so we can ignore that part of the function. The derivative of a term with a coefficient of 1 (such as 2x in this case) is just the coefficient times the variable (in this case, 2x), so the derivative of 2x is 2.

The derivative of a term with a coefficient other than 1 (such as 3x^2 in this case) is the coefficient times the derivative of the term with the coefficient of 1 (in this case, 3 times x), so the derivative of 3x^2 is 3x.

Now that we have found the derivative of each term in the function, we can add them up to get the derivative of the whole function:

f'(x) = 3x + 2 = 6x + 2

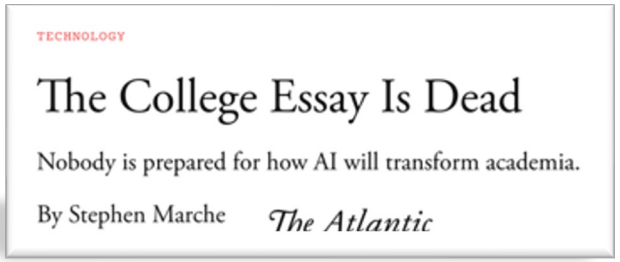
This is the final answer. The derivative of the function f(x) = 3x^2 + 2x - 1 is f'(x) = 6x + 2.



Socratic critique of technology

Such tools may impart "not truth but only the semblance of truth" and may appear to be omniscient and will generally know nothing."

The implications of all this for education/ university


TECHNOLOGY

The College Essay Is Dead

Nobody is prepared for how AI will transform academia.

By Stephen Marche *The Atlantic*

MOTHERBOARD
TECH BY VICE

ChatGPT Is Passing the Tests Required for Medical Licenses and Business Degrees

ChatGPT passes exams from law and business schools


By Samantha Murphy Kelly, CNN Business
Updated 1:35 PM EST, Thu January 26, 2023

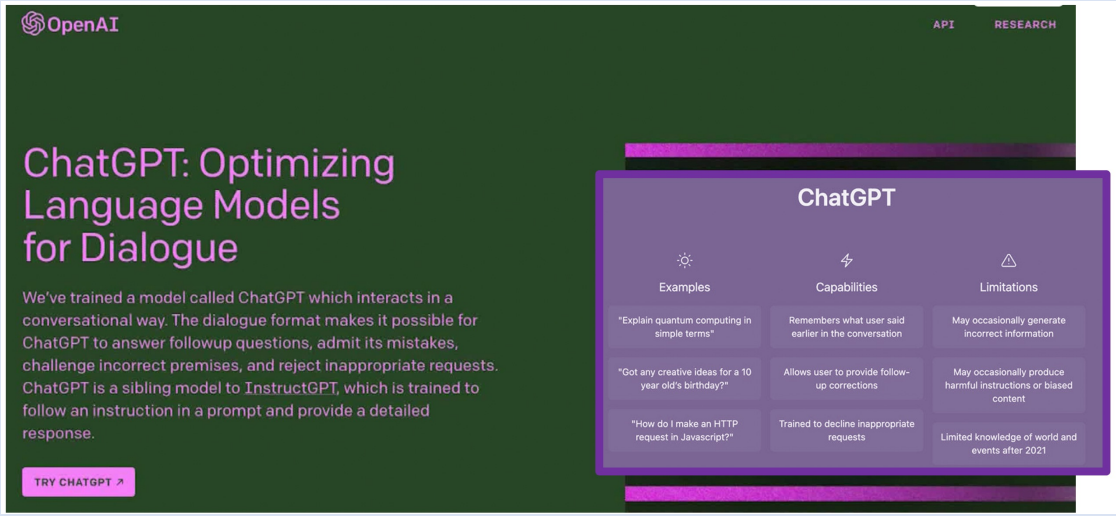
ChatGPT passes Wharton Business School test: research paper

Engineering Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education

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IEEE EDUCON 2023
IEEE Global Engineering Education Conference

Questions/ Discussion Mini-Break



Engineering Education in the Era of ChatGPT: Promise and Pitfalls of Generative AI for Education

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2

A Critical Look at ChatGPT

Yann Le Cun on LLMs



Yann LeCun
@ylecun

My unwavering opinion on current (auto-regressive) LLMs
1. They are useful as writing aids.
2. They are "reactive" & don't plan nor reason.
3. They make stuff up or retrieve stuff approximately.
4. That can be mitigated but not fixed by human feedback.
5. Better systems will come

4:02 PM · Feb 13, 2023 · **723.8K** Views

6. Current LLMs should be used as writing aids, not much more.
7. Marrying them with tools such as search engines is highly non trivial.
8. There *will* be better systems that are factual, non toxic, and controllable. They just won't be auto-regressive LLMs.

Is ChatGPT reliable and trustworthy?

ChatGPT is "high-tech plagiarism" for lazy learners, says Noam Chomsky

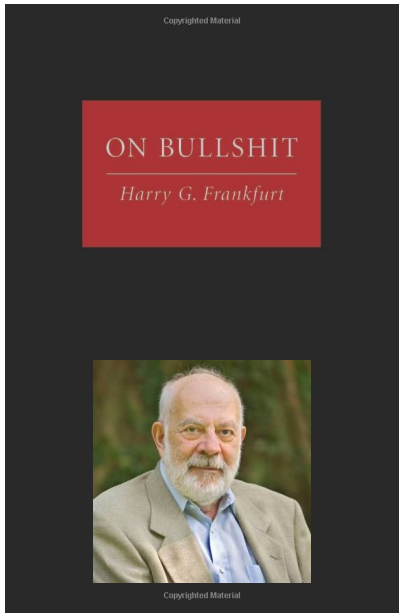




Castles in the Air



ChatGPT in less flattering terms



ChatGPT is a bullshit generator. But it can still be amazingly useful

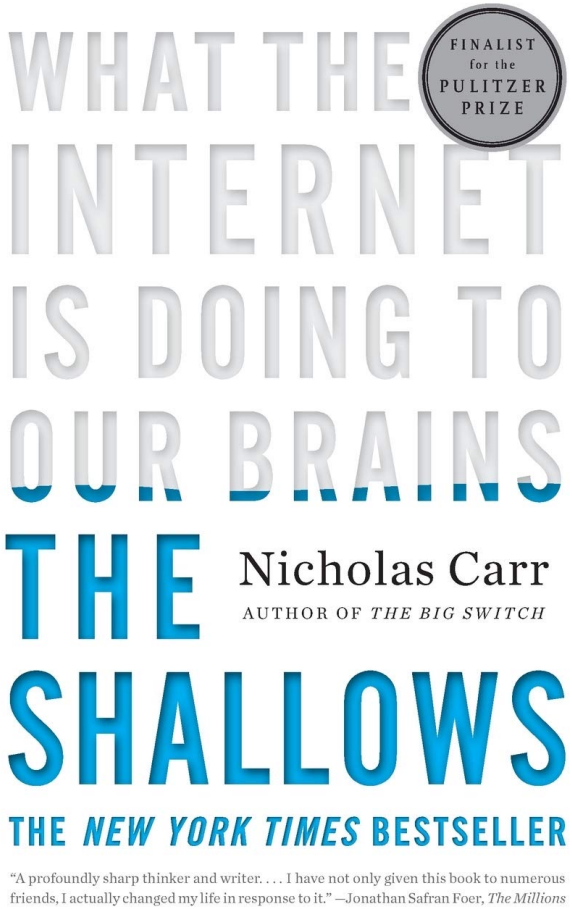


Arvind Narayanan and Sayash Kapoor
Dec 6, 2022

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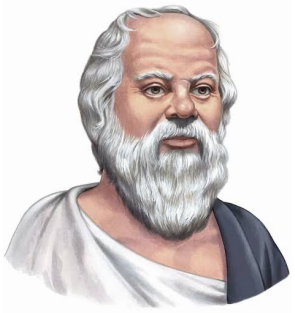
The philosopher Harry Frankfurt defined bullshit as speech that is intended to persuade without regard for the truth.

The risk of outsourcing thinking to technology



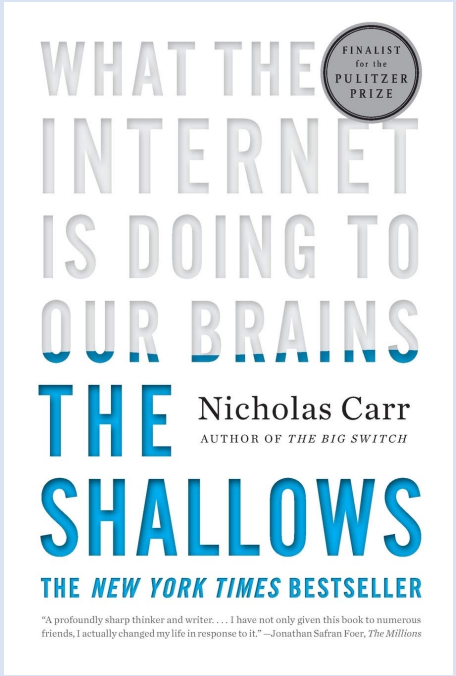
Plato's Argument Against Writing

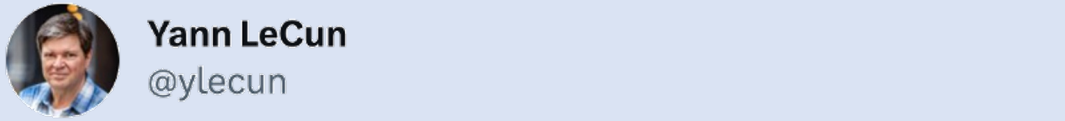
"If men learn this, it will implant forgetfulness in their souls. ***They will cease to exercise memory because they rely on that which is written, calling things to remembrance no longer from within themselves, but by means of external marks.***"



Such tools impart "not truth but only the semblance of truth" and that those who adopt it will appear to omniscient and will generally know nothing."

Questions/ Discussion Mini-Break







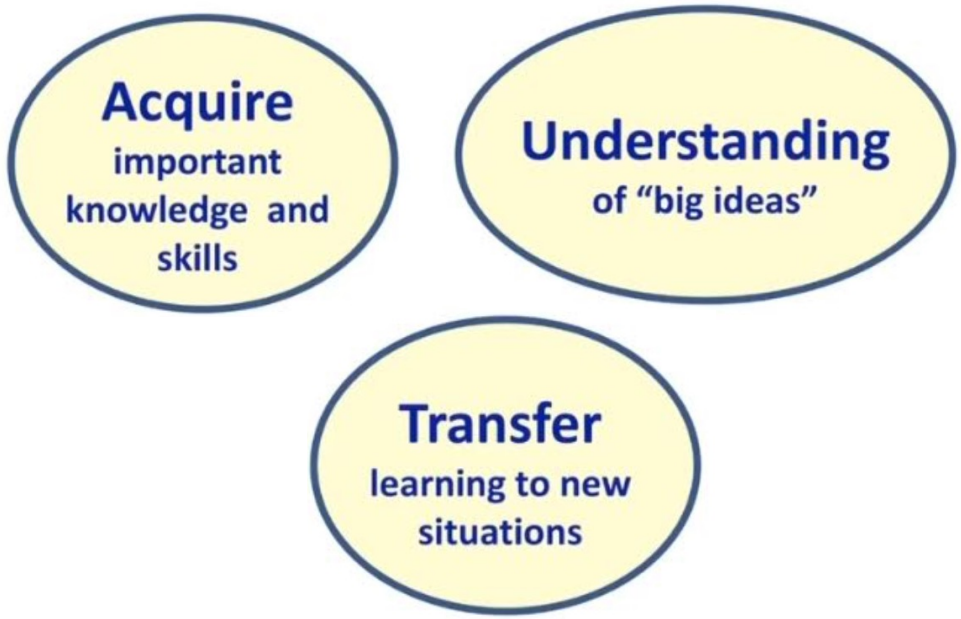


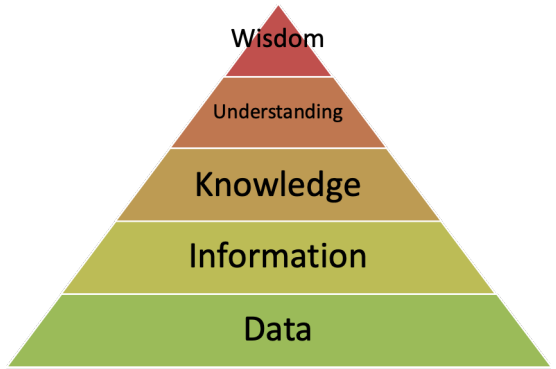
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A Deeper Look at:

What is Education?

Different types of learning goals





Who's educated?
What's education?

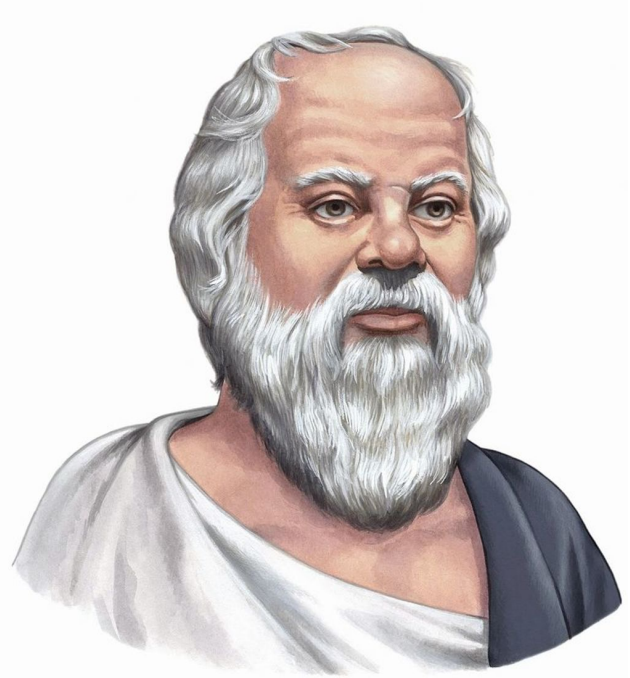
Is it the same as having a degree?

Is it the same as having good grades?

Is it the same as knowing/providing information?

Two vignettes on learning and education

SOCRATES AND THE SOPHISTS



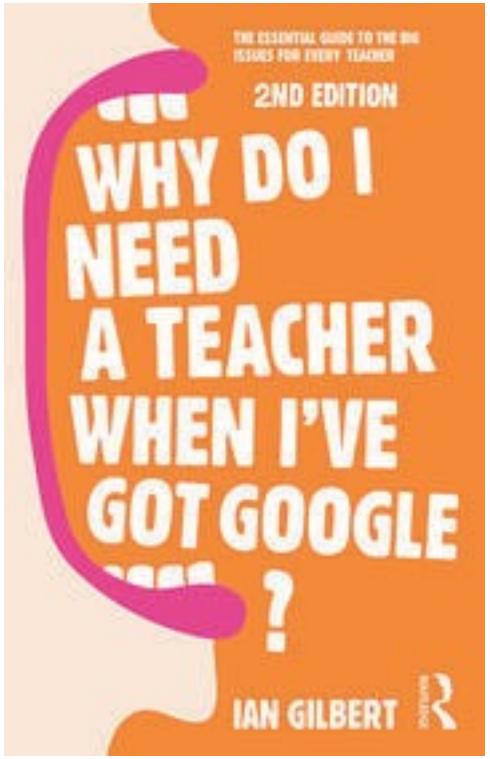
Appearences can be deceiving

AL-GHAZALI AND THE ROBBERS



Education and knowledge is inside you

Can learning be automatized and outsourced?



You need to train your mind by doing this and go through the process to self-discover or make your concept your own.

Information Technology and Educational Amnesia

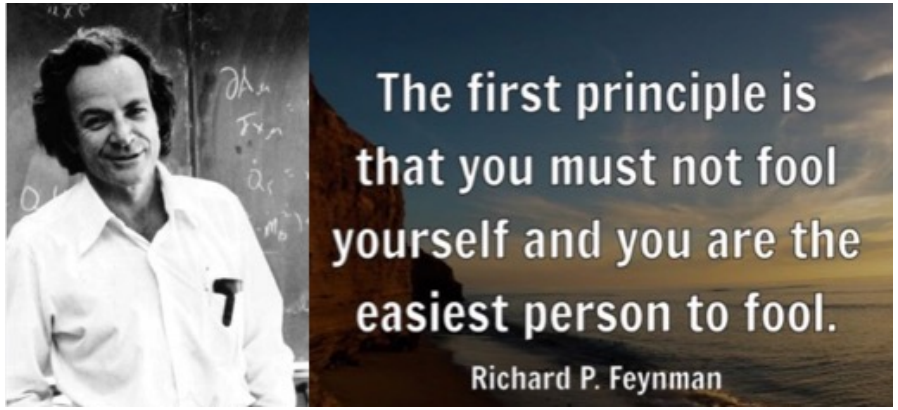
LANGDON WINNER
Rensselaer Polytechnic Institute, Troy, New York, USA

No matter how wonderful the technology (TV, VCR, Internet, Google, ChatGPT), technology alone is not sufficient. You need analog complements.

Technology can amplify human intent and capabilities BUT not substitute for them.

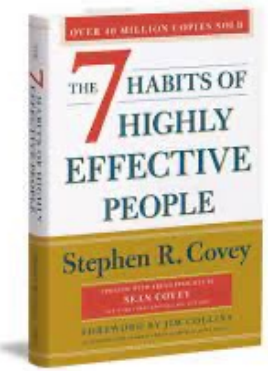
Law of the Harvest: Beware of Shortcuts

Time needed for meaning-making and Understanding:
(Otherwise, we could give all the answers/ info at the start)

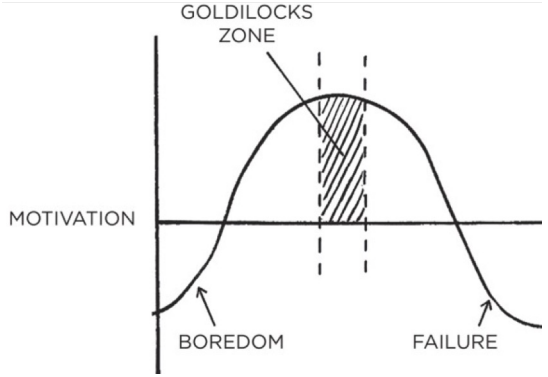


Cramming
Outsourcing
Copying





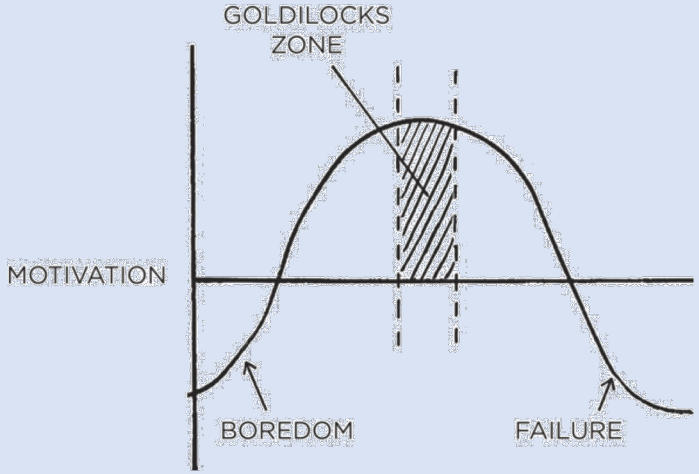
Theory of
"Desirable Difficulties"
[Bjork]

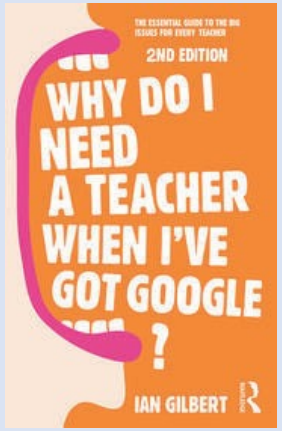


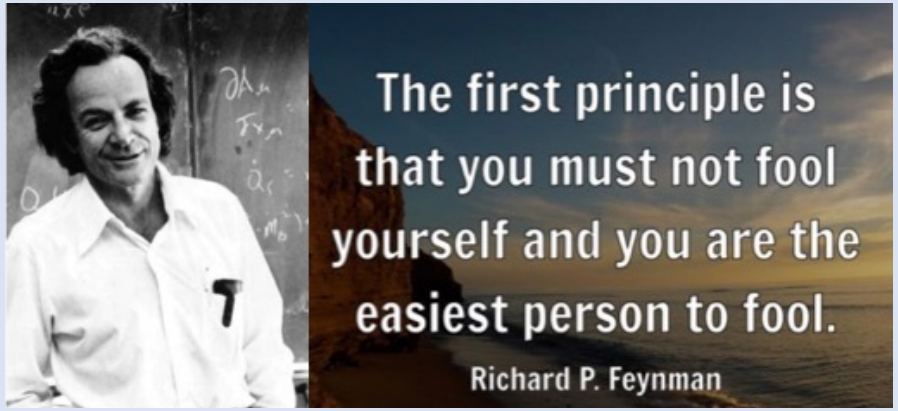
Questions/ Discussion Mini-Break

Information Technology and Educational Amnesia

LANGDON WINNER
Rensselaer Polytechnic Institute, Troy, New York, USA







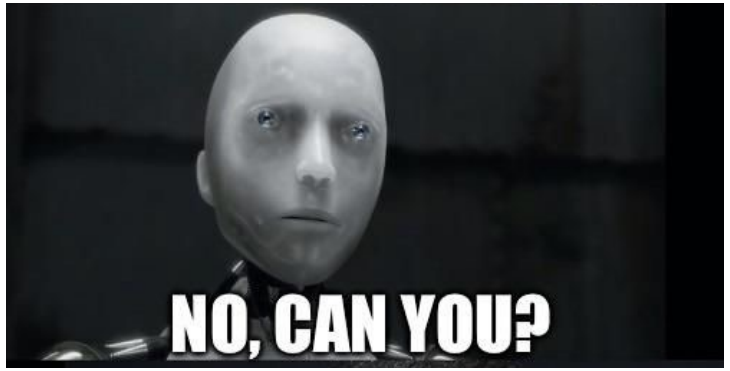
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How students can use ChatGPT: a primer

"Efficient learning means refusing passivity, engaging, exploring, and actively generating hypotheses and testing them on the outside world."—S. Dehaene.

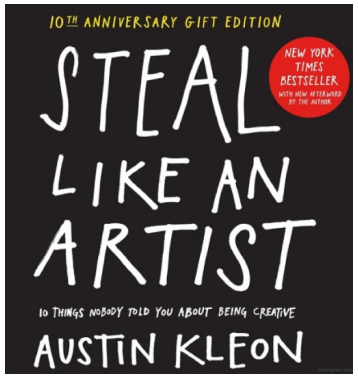
Plagiarism vs. Inspiration vs. Stealing

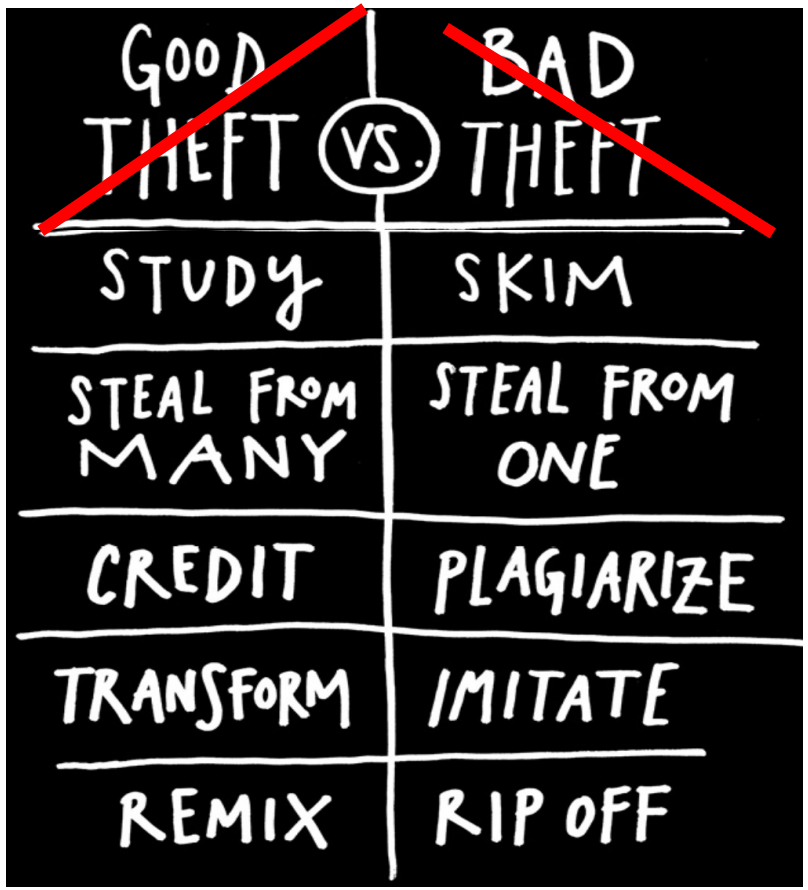




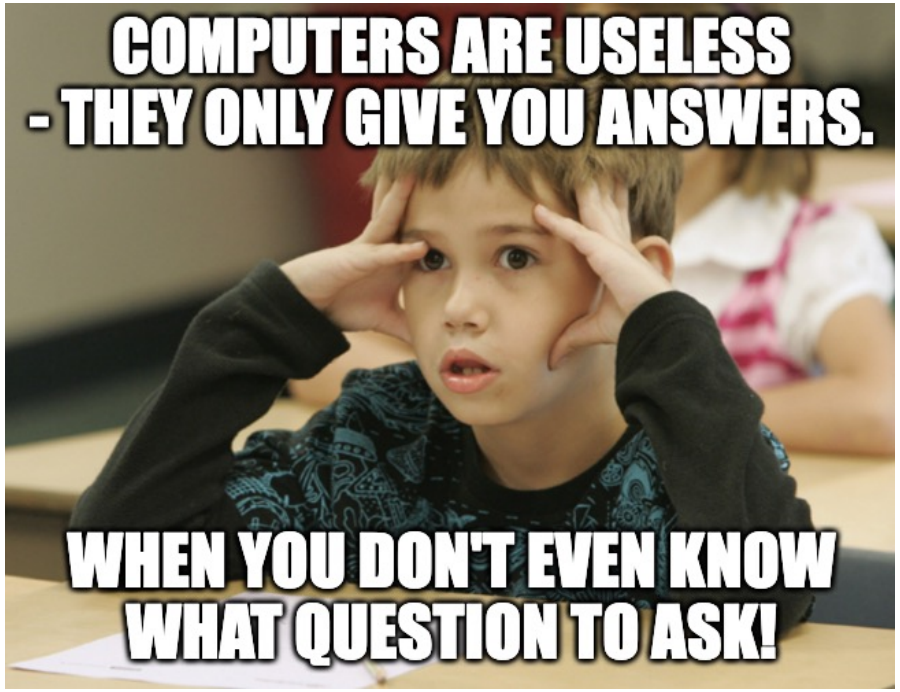


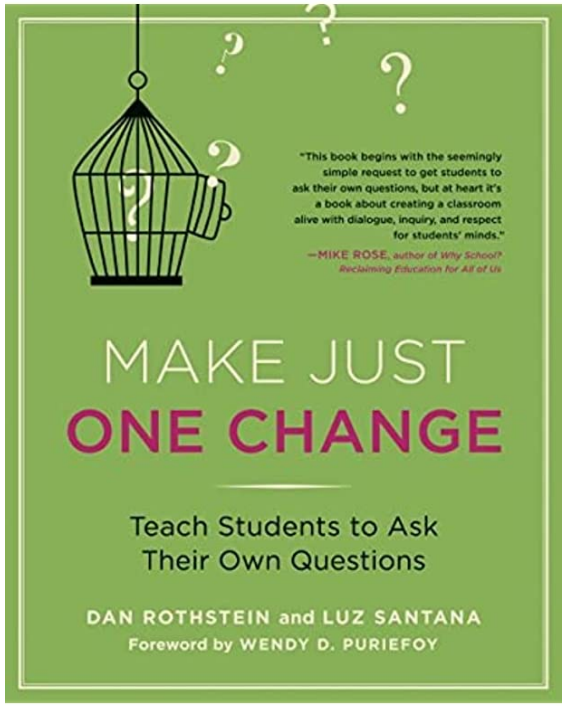
INSPIRATION PLAGIARISM





Enhance your skill of asking questions

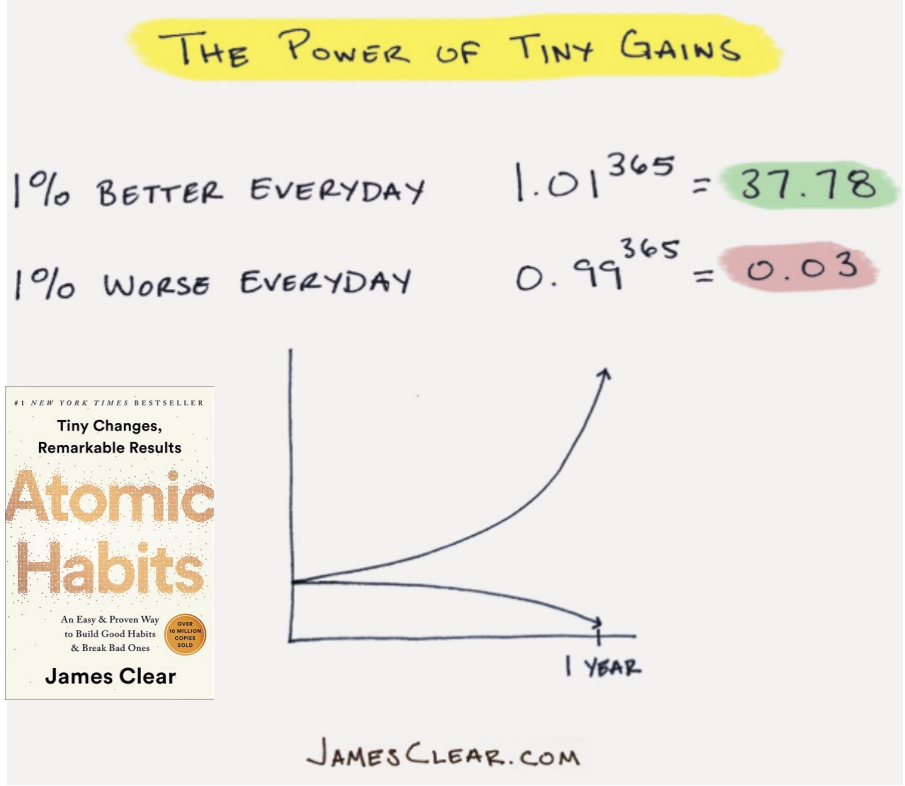




Because a modern education seeks to engage students in meaning making and to develop self-directed learners, students should be encouraged to develop their own questions—and to pursue the answers through active inquiry (i.e., to become researchers)

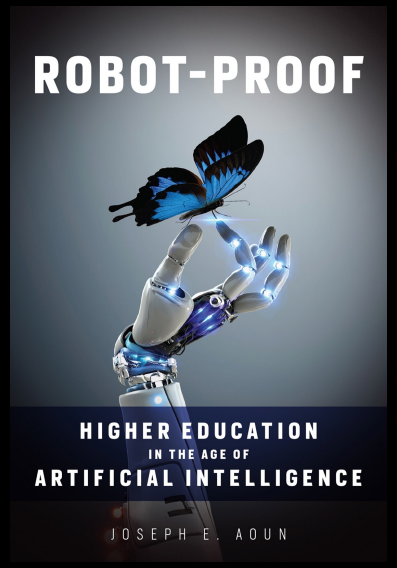
Lifelong Learning … Investing in Yourself

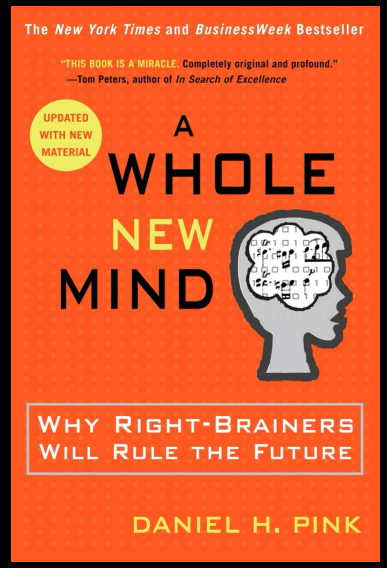
When students acquire meta-skills such as learning how to learn, read, write, they bring this skill to every course and reap its ***exponential benefits****—this effect is sometimes called the "**Matthew Effect**", or the rich get richer and the poor get poorer effect.*



Thus investing in the basics provides compound interest over time

Becoming AI/Bot Proof





How should we live in times of exponential technologies?

In a Markdown table what are 10 jobs that can't be taken by AI and why

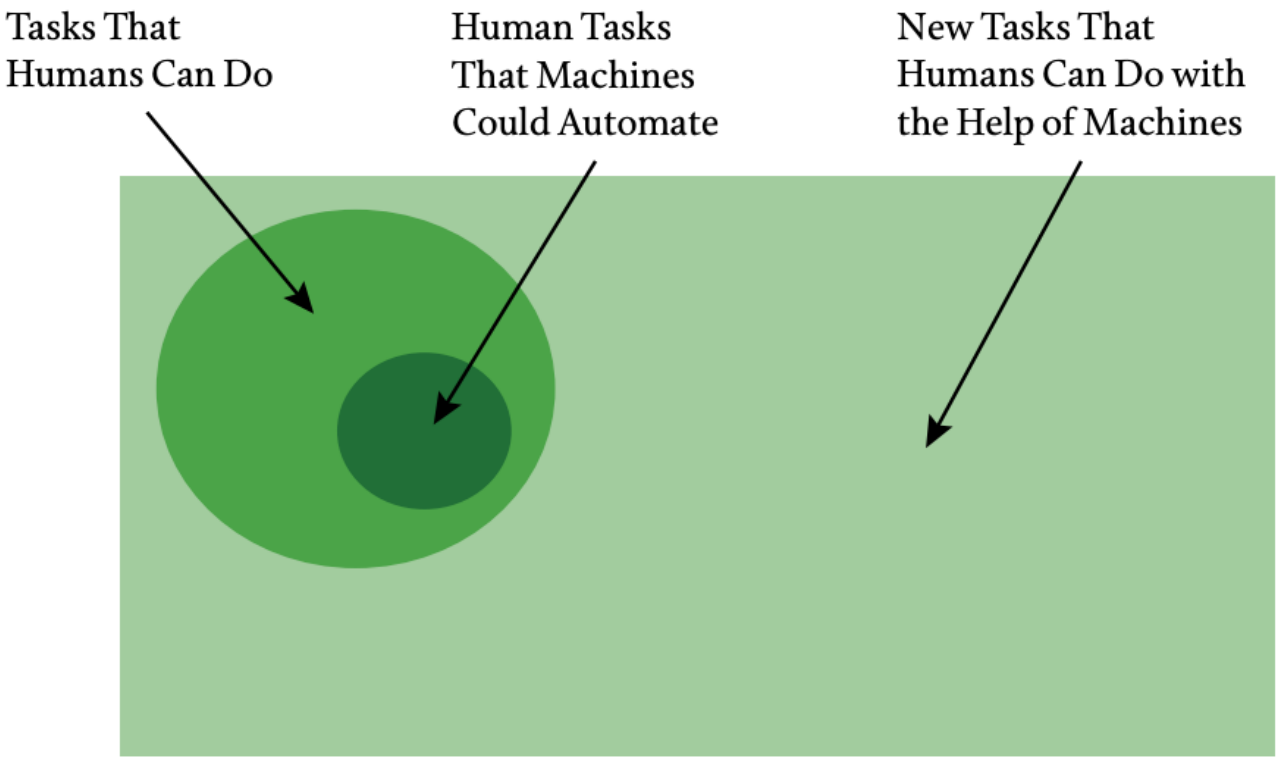
Job	Why it can't be taken by AI
Psychologist	AI lacks empathy and human touch, which is essential in understanding human emotions and behavior
Artist	Creativity is a uniquely human trait that AI hasn't been able to replicate
Social Worker	Addressing complex social issues and building relationships with clients requires a human touch that AI cannot provide
	navigate without human guidance
Entrepreneur	Starting and building a successful business requires human creativity, risk-taking, and adaptability
Doctor	AI can assist with diagnosis and treatment recommendations, but it cannot replace the empathy and communication skills required in patient care
Politician	Decision-making in politics involves value judgments and ethical considerations that require human reasoning and judgment

The risk of being outdated, redundant, and jobless

- Some people will be made jobless by AI (due to automation)

- Most people will be outdone by humans +AI.

How Can Use ChatGPT as the bicycle of Human Intelligence?





AI for Education should be:
(1) Human-Centered; and
(2) based on Intelligence Augmentation

"Laws" of Human Intelligence Augmentation

Law 1

if the machine is not there .. be at least as good as you were before. don't outsource yourself

Law 2

if the machine is there, be no worse than the machine.
(you can use the machine to be no worse)

Law 3

if the machine is there.. and if you don't want to be made irrelevant, strive to maintain your competitive advantage over bots/algorithms/machines.

Fair Use Tips and Guidelines

Inquire about "AI Tools Fair Usage Policy"

الحلال بين والحرام بين وبينهما امور مشتبهات

You should not steal/ copy without attribution.

You should not deceive.

Cite and credit AI

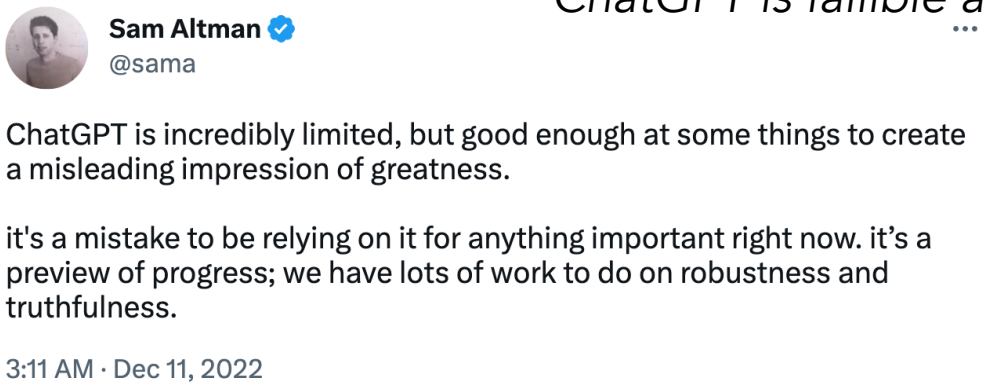
Cite the appropriate sources to support claims in the answer that you get from AI (otherwise be skeptical of the claims)

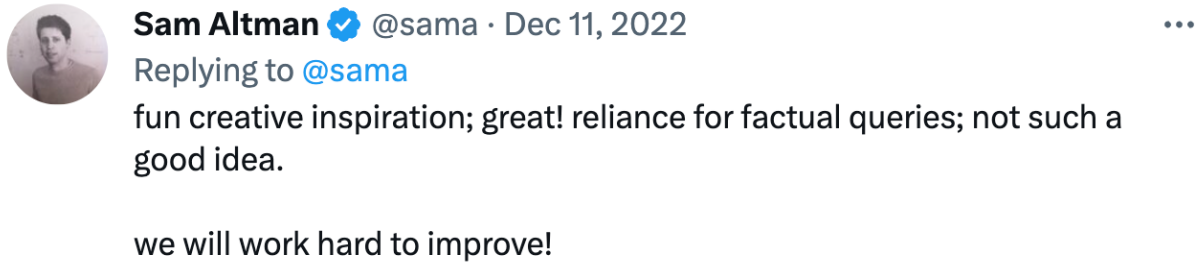
Use AI as a tool to learn, not just to produce content.

When in doubt, check with your instructor about acceptable AI usage for your courses

Caveat Emptor: Let the Buyer Be Aware

ChatGPT is fallible and it makes errors and even hallucinates.





Let's say it's highly accurate (e.g., 90% of the time, it is factual).

Can we use it for critical tasks?

***The problem is that we do not know which 90%**, and worse, ChatGPT doesn't know either!"*

- *Importance of Fact-Checking ChatGPT's Output:* Don't trust anything it says. If it gives you a number or fact, assume it is wrong unless you either know the answer or can check with another source.
- You, not ChatGPT, is responsible for any errors/ omissions.

Thank you!
Questions/ Discussion

Engineering Education, Moving into 2020s:
Essential Competencies for Effective 21st Century Electrical & Computer Engineers

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